Statistical Inference for Series Systems from Masked Failure Time Data: The Exponential Case

Alexander Towell
atowell@siue.edu

Abstract

We derive closed-form statistical inference for m-component series systems with exponentially distributed lifetimes when the failure cause is only partially observed. Under a uniform masking model—a tractable specialization of the general C1–C2–C3 masked-data likelihood framework—we characterize the score equations, Fisher information matrix, and sufficient statistics for arbitrary system dimension m and masking cardinality w, and derive a closed-form MLE for the maximally uninformative identifiable case $w = m - 1$. In this case the score equations linearize and the MLE admits a closed-form solution for general m. We prove that uniform masking maximizes the conditional entropy of the candidate set among all symmetric (C2) masking models, and that diagnostic informativeness—measured by the mutual information $\mathrm{I}(K; C)$—decreases monotonically with w via a data processing inequality. The Fisher information matrix quantifies how diagnostic precision (smaller w) translates into statistical efficiency, and its singularity at $w = m$ reflects the non-identifiability of completely masked data. Monte Carlo simulations confirm that the asymptotic covariance matches empirical behavior at rate $O(n^{-1/2})$ across diverse parameter configurations. The three-component system with $w = 2$ is developed in detail as an illustration.

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1 Introduction

Series systems are fundamental in reliability engineering: the system fails whenever any single component fails. In many practical situations, the exact cause of system failure cannot be determined with certainty, but can be narrowed to a subset of components. This partial information is known as *masked* failure data. For example, in electronic systems a diagnostic test might isolate a failure to one of two circuit boards without determining which board actually failed; in medical devices, system failure might be attributed to a subset of components based on failure mode symptoms; or in aerospace systems, post-failure analysis might narrow the cause to a specific subsystem but cannot identify the individual component responsible.

The problem arises in two common scenarios: (1) diagnostic limitations where identifying the exact failed component requires destructive testing or prohibitive expense, and (2) field data collection where only partial diagnostic information is recorded. Despite this incomplete information, engineers must estimate component-level reliability parameters to make design improvements, optimize maintenance schedules, and predict system lifetime distributions.

1.1 Related Work

The analysis of masked failure data has its roots in competing risks theory. Cox [4] introduced the latent failure time model for analyzing exponentially distributed lifetimes with multiple failure types, establishing the foundational framework for modeling situations where multiple competing causes may lead to system failure. This work demonstrated that under exponential assumptions, the failed component identity and system lifetime can be independent random variables, a property that proves crucial for the analysis in this paper.

Miyakawa [15] pioneered the application of competing risks methods specifically to masked system failure data, deriving closed-form maximum likelihood estimates for a two-component series system with exponential component lifetimes. Usher and Hodgson [20] extended this approach, introducing general maximum likelihood methods for estimating component reliability from masked system life-test data and providing computational procedures for series systems. Lin et al. [12] further developed exact maximum likelihood estimation procedures for three-component systems under more general masking scenarios.

Dinse [6] developed nonparametric methods for partially-complete time and type of failure data, introducing an iterative algorithm yielding distribution-free estimates that converge to maximum likelihood solutions. This work established that useful inference is possible even when the failure cause is only partially observed.

The 1990s saw significant development of Bayesian approaches to masked data. Reiser et al. [17] introduced Bayesian inference methods for masked system lifetime data, while Guttman et al. [10] addressed dependent masking scenarios where the probability of masking depends on the true cause of failure, developing Bayesian methodology for two-component systems. These Bayesian approaches provided posterior distributions for component reliabilities but did not derive closed-form expressions for frequentist estimators or their asymptotic properties.

Flehinger et al. [7, 8] contributed to the theoretical foundations by studying survival analysis with competing risks and masked failure causes, developing parametric models that accommodate general patterns of missing failure types. Sarhan [18] provided both maximum likelihood and Bayes procedures for estimating component reliabilities in series systems with exponentially distributed lifetimes, focusing on practical implementation issues.

Craiu and Duchesne [5] developed EM-based inference for piecewise-constant cause-specific hazards with masked failure causes, providing both E- and M-steps in closed form—a semiparametric counterpart to the fully parametric closed-form results derived here.

More recent work has expanded the scope to include interval-censored data [9], Bayesian modeling frameworks [11], nonparametric Bayesian methods with dependent competing risks [14], and applications to specific engineering contexts. Identifiability—the structural question underlying all masked-data analysis—has been revisited through phase-type models by Lindqvist [13], who provides identifiable parameterizations for competing risks with multiple absorbing states. The competing risks framework continues to provide a natural setting for masked data analysis [1], connecting reliability engineering with survival analysis and biostatistics; see Monterrubio-Gómez et al. [16] for a recent comprehensive review. Towell [19] develops a distribution-agnostic likelihood framework under three conditions on the masking mechanism (C1–C2–C3) that enable the unknown masking distribution to be eliminated from the likelihood, and establishes formal identifiability conditions based on the combinatorial structure of the candidate sets. The present paper specializes that framework to exponential lifetimes under a stronger uniform masking assumption (see Remark 2.8 below), which enables the closed-form results that are the focus of this work.

Despite this substantial body of work, *closed-form analytical results remain limited*. Specifically:

- The Fisher information matrix for masked system data has not been derived in closed form for general series systems, even under exponential assumptions
- Asymptotic properties of maximum likelihood estimators have not been fully characterized without numerical computation
- Minimal sufficient statistics have not been identified for general masking patterns
- Most existing methods rely on numerical optimization or EM algorithms without explicit characterization of estimator variance and covariance structure

This paper addresses these gaps by providing complete analytical results for the exponential case. The exponential distribution is a cornerstone of reliability theory [2], justified in several contexts: systems subject to random external shocks, systems with constant hazard rates, and early-life failure analysis where wear-out has not yet begun. The exponential assumption also serves as a limiting case for other distributions and provides a baseline for comparison with more complex models. More importantly, the exponential case admits closed-form solutions that provide direct insight into the information structure of masked failure data.

1.2 Contributions

This paper provides a complete analytical treatment of maximum likelihood estimation for general m-component series systems with masked failure data under exponential component lifetimes. While closed-form MLEs have been derived for specific small systems [15, 12], complete results—including the Fisher information matrix, sufficient statistics, and asymptotic theory—have not previously been obtained for arbitrary m. Our main contributions are:

1. **Explicit Fisher information matrix**: We derive a closed-form expression for the Fisher information matrix under arbitrary masking

patterns, enabling direct computation of asymptotic variances without numerical differentiation or Monte Carlo simulation

2. **Sufficient statistics**: The mean system lifetime and candidate set frequency vector constitute sufficient statistics, reducing the data to $1 + \binom{m}{w}$ real numbers

3. **Closed-form MLE for $w = m-1$ masking**: For masking cardinality $w = m-1$ (where each candidate set excludes exactly one component), we derive an explicit closed-form solution to the likelihood equations for arbitrary m, eliminating the need for numerical optimization. This extends prior closed-form results for two- and three-component systems [15, 12] to arbitrary system dimension m. The three-component case ($m = 3$, $w = 2$) is developed in detail

4. **Asymptotic distribution theory**: We characterize the asymptotic sampling distribution of the MLE and provide Wald-type confidence intervals with explicit formulas for coverage

5. **Numerical validation**: We provide Monte Carlo evidence confirming the theoretical asymptotic covariance matches empirical behavior for finite samples

The closed-form results for $w = m - 1$ are particularly significant because this is the *maximally symmetric identifiable* masking configuration under uniform masking. The masking cardinality ranges from $w = 1$ (exact cause identification, no masking) through $w = m$ (complete masking, non-identifiable). Each candidate set of size w excludes $m - w$ components from suspicion, so under uniform masking, larger w means less diagnostic information per observation. At $w = m - 1$, each observation excludes only a single component, retaining the minimum diagnostic information needed for identifiability under uniform masking. The $w = m-1$ case is also where the score equations linearize, enabling the closed-form MLE that is the central result of this paper.

1.3 Paper organization

Section 2 introduces the probabilistic model for series systems with masked failures. Section 3 develops the likelihood and Fisher information for general parametric families. Section 4 presents the main results for exponentially distributed components, including the MLE, information matrix, and sufficient statistics. Section 5 derives the closed-form MLE for $w = m - 1$ and develops the three-component case in detail. Section 6 concludes with discussion. Appendix A provides a Newton–Raphson algorithm for the general-w case.

2 Probabilistic Model

2.1 Series System Lifetime

Consider a system composed of m components. Component j has a random lifetime $\mathrm{T}_j > 0$ for $j = 1, \ldots, m$. We make the following assumptions:

Assumption 2.1. The component lifetimes $\mathrm{T}_1, \ldots, \mathrm{T}_m$ are mutually independent.

Assumption 2.2. The system operates if and only if all components are functioning (series configuration).

Under these assumptions, the system lifetime is:

$$\mathrm{S} = \min(\mathrm{T}_1, \ldots, \mathrm{T}_m) \tag{1}$$

Let $F_j(t)$ and $f_j(t)$ denote the CDF and PDF of component j. Define the reliability function $R_j(t) = 1 - F_j(t) = \mathrm{P}\{\mathrm{T}_j > t\}$.

The system reliability function is:

$$R_{\mathrm{S}}(t) = \prod_{j=1}^{m} R_j(t) \tag{2}$$

The system PDF is:

$$f_{\mathrm{S}}(t) = \sum_{j=1}^{m} \left(f_j(t) \prod_{\substack{k=1 \\ k \neq j}}^{m} R_k(t) \right) \tag{3}$$

2.2 Masked Component Failures

When the system fails at time t, exactly one component caused the failure. Let $\mathrm{K} \in \{1, \ldots, m\}$ denote the failed component. In many applications, K cannot be observed directly. Instead, we observe a *candidate set* $\mathrm{C} \subseteq \{1, \ldots, m\}$ that contains the failed component but does not uniquely identify it.

Definition 2.3. Under *uniform masking* with cardinality w, given that component k failed, the observed candidate set C is a uniformly random subset of size w containing k. That is,

$$\mathrm{P}\{\mathrm{C} = c | \mathrm{K} = k\} = \begin{cases} \frac{1}{\binom{m-1}{w-1}} & \text{if } k \in c \text{ and } |c| = w \\ 0 & \text{otherwise} \end{cases} \tag{4}$$

Remark 2.4 (Interpretation and Implications)*.* The uniform masking model has two key components:

1. The failed component k is *always* included in the candidate set ($\mathrm{P}\{k \in \mathrm{C} | \mathrm{K} = k\} = 1$)

2. Each non-failed component $j \neq k$ is included in C with *equal probability* $\frac{w-1}{m-1}$, independently

This is a **strong assumption** that is generally *not satisfied in practice.* In real diagnostic scenarios:

- Components with similar failure modes may be systematically grouped together

- Accessible components may be more likely to be identified than inaccessible ones

- Expensive tests may create diagnostic biases

- Prior failure history may influence which components are suspected

Remark 2.5 (Justification for Uniform Masking)*.* We adopt the uniform masking assumption because it enables *analytical derivation* of the score equations, Fisher information matrix, and sufficient statistics—and, for $w = m - 1$, a closed-form MLE. These analytical results provide:

- Direct insight into the information structure of masked failure data
- Closed-form asymptotic variance formulas (no Monte Carlo needed)
- Baseline performance bounds for comparison with more complex masking models
- Computational efficiency (no numerical optimization needed for $m = 3$, $w = 2$)

The uniform masking model is most appropriate when:

- The diagnostic process randomly samples components for inspection
- The masking mechanism is designed to be unbiased (e.g., automated random selection)
- No prior information about component reliabilities is used during diagnosis

Despite its limitations, the uniform masking model with $w = m - 1$ serves as a natural *reference model* for masked failure data. Uniform masking minimizes diagnostic information $\mathrm{I}(\mathrm{K}; \mathrm{C})$ among identifiable masking mechanisms at each cardinality (Proposition 2.6), though note that diagnostic information about the failed component and Fisher information about the parameters $\boldsymbol{\lambda}$ are distinct quantities (see the remark following Proposition 2.6). The analytical tractability of uniform masking arises from its maximal symmetry—uniform inclusion probability for non-failed components—analogous to how the exponential distribution's tractability arises from the constant hazard rate.

The following result makes the "maximum-entropy–like" claim precise. It is distribution-free: it depends only on the masking mechanism, not on the component lifetime distributions.

Proposition 2.6 (Diagnostic Informativeness)**.** *Let* $\mathrm{I}(\mathrm{K}; \mathrm{C}_w)$ *denote the mutual information between the failed component* K *and the candidate set* C *under uniform masking with cardinality* w*.*

1. ***Maximum conditional entropy.*** *For fixed* w*, among all masking models satisfying* C1 *(containment) and* C2 *(symmetry), uniform masking maximizes the conditional entropy* $\mathrm{H}(\mathrm{C}|\mathrm{K})$*.*

2. ***Monotone information loss.*** *Under uniform masking,* $\mathrm{I}(\mathrm{K};\mathrm{C}_w)$ *is strictly decreasing in* w*. Consequently,* $w = m-1$ *minimizes the diagnostic information among all identifiable* ($w < m$) *uniform masking configurations.*

Proof. For part (1): under C2, $\mathrm{P}\{\mathrm{C} = c \mid \mathrm{K} = k\} = \beta_c$ for all $k \in c$, with $\sum_{c \ni k} \beta_c = 1$ for each k. The conditional entropy decomposes as $\mathrm{H}(\mathrm{C}|\mathrm{K}) = \sum_k \mathrm{P}\{\mathrm{K} = k\}\,\mathrm{H}(\mathrm{C}|\mathrm{K} = k)$, where each $\mathrm{H}(\mathrm{C}|\mathrm{K} = k) = -\sum_{c \ni k} \beta_c \ln \beta_c$ is maximized by the uniform distribution $\beta_c = 1/\binom{m-1}{w-1}$. Since this choice satisfies the constraint for all k simultaneously, it maximizes every term.

For part (2): given C_w (a uniform random size-w set containing K), define $\mathrm{C}_{w+1} = \mathrm{C}_w \cup \{J\}$ where J is drawn uniformly from $\{1, \ldots, m\} \setminus \mathrm{C}_w$. The number of size-$w$ subsets of a given size-$(w+1)$ set c' that contain K is w, giving

$$\mathrm{P}\{\mathrm{C}_{w+1} = c' \mid \mathrm{K} = k\} = \frac{w}{\binom{m-1}{w-1}(m-w)} = \frac{1}{\binom{m-1}{w}},$$

which is the uniform masking distribution with cardinality $w+1$. This establishes the Markov chain $\mathrm{K} \to \mathrm{C}_w \to \mathrm{C}_{w+1}$, and the data processing inequality yields $\mathrm{I}(\mathrm{K};\mathrm{C}_{w+1}) \le \mathrm{I}(\mathrm{K};\mathrm{C}_w)$. Strict inequality holds because the channel $\mathrm{C}_w \to \mathrm{C}_{w+1}$ is not invertible. □

Remark 2.7. In the symmetric case $\lambda_1^\star = \cdots = \lambda_m^\star$, the mutual information has the explicit form $\mathrm{I}(\mathrm{K};\mathrm{C}_w) = \ln(m/w)$, which ranges from $\ln m$ (exact identification, $w = 1$) through $\ln(m/(m-1))$ (maximum masking with identifiability, $w = m-1$) to 0 (complete masking, $w = m$).

Note that $\mathrm{I}(\mathrm{K};\mathrm{C})$ measures how well the candidate set identifies the *failed component*, not how precisely the *parameters* $\boldsymbol{\lambda}^\star$ can be estimated. The Fisher information matrix (Proposition 4.6) depends on the full masking mechanism, not only on $\mathrm{I}(\mathrm{K};\mathrm{C})$. Indeed, a deterministic masking rule at $w = m-1$ that maps each failure cause to a unique candidate set recovers the same FIM as complete identification ($w = 1$), despite having cardinality $w = m-1$ (Proposition 4.7). Uniform masking, by contrast, provides no preferential information about component identity; the resulting closed-form FIM serves as an analytical benchmark under maximal masking symmetry, not a worst-case bound.

Remark 2.8 (Relationship to the C1–C2–C3 Framework)*.* The uniform masking model (Definition 2.3) implies the three standard conditions used in the general masked-data likelihood literature [19]:

C1. The candidate set contains the failed component: $\mathrm{P}\{\mathrm{K} \in \mathrm{C}\} = 1$.

C2. The candidate set probability is symmetric across components within the set: $\mathrm{P}\{\mathrm{C} = c \mid \mathrm{K} = j\} = \mathrm{P}\{\mathrm{C} = c \mid \mathrm{K} = j'\}$ for all $j, j' \in c$.

C3. Masking probabilities do not depend on the system parameter Θ.

The uniform masking assumption is *strictly stronger* than C1–C2–C3 in two respects:

1. **Fixed cardinality.** We require $|\mathrm{C}| = w$ for every observation. Under C1–C2–C3, the candidate set size may vary: different observations may have $|\mathrm{C}_i| = 2, 3, \ldots$ The fixed cardinality is a design constraint (e.g., the diagnostic always reports a candidate set of a prespecified size).

2. **Equal inclusion probability.** Each non-failed component is included in C with the same probability $(w-1)/(m-1)$, independently. C2 requires only that, *given a particular candidate set* c, the probability of observing c does not depend on which member of c failed. Uniform masking goes further: all $\binom{m-1}{w-1}$ candidate sets containing the failed component are equally likely.

Under C1–C2–C3 alone, the masking probability $\beta_i = \mathrm{P}\{\mathrm{C}_i = c_i \mid \mathrm{S}_i = t_i, \mathrm{K}_i = j\}$ is an unknown nuisance parameter that factors out of the likelihood (C3) and thus does not affect the MLE. The point estimates are therefore identical. However, the closed-form Fisher information matrix and sufficient statistics derived in this paper *do* require knowledge of the masking distribution: the FIM entries involve expected candidate set frequencies, which under uniform masking are determined by the known probability $1/\binom{m-1}{w-1}$. Under C1–C2–C3 with unknown β_i, one obtains the same MLE but cannot compute asymptotic variances in closed form without additional modeling of the masking mechanism.

Connection to identifiability. The identifiability theory for masked data [19] shows that parameters are identifiable when the masking mechanism is *separating*: for every pair of components $j \neq j'$, there exists a candidate set c containing j but not j'. Under uniform masking with cardinality $w < m$, every candidate set of size w occurs with positive probability, and in particular the set $\{1, \ldots, m\} \setminus \{j'\}$ separates j from j' whenever $w \leq m-1$. Thus uniform masking with $w \leq m-1$ is automatically separating, guaranteeing identifiability. Conversely, when $w = m$ every candidate set is $\{1, \ldots, m\}$ and all component pairs are diagnostically confounded, explaining the Fisher information singularity at $w = m$ (see condition R2 preceding Theorem 4.10).

Assumption 2.9. The masking mechanism (which components are included in C) is independent of the system lifetime S and depends only on the failed component K.

This assumption implies that the masking quality does not depend on when the failure occurred, only on which component failed and the diagnostic capabilities of the inspection process.

Definition 2.10. A *masked system failure time* is a pair (t, C) where t is the observed system failure time and C is the candidate set. Let $\mathrm{W} = |\mathrm{C}|$ denote the *masking cardinality*. When $\mathrm{W} = w$ is fixed by experimental design, we condition on w throughout.

2.3 Parametric Families

We assume component j has a lifetime distribution from a parametric family indexed by parameter $\theta_j^\star$. The system parameter is $\Theta^\star = (\theta_1^\star, \ldots, \theta_m^\star)$.

Our data consists of a random sample of n independent masked system failure times:

$$\mathbf{M}_n = \{(\mathrm{t}_1, \mathrm{C}_1), \ldots, (\mathrm{t}_n, \mathrm{C}_n)\} \tag{5}$$

where we condition on fixed cardinality w for simplicity.

3 Likelihood and Fisher Information

3.1 Likelihood Function

Given the model assumptions, the conditional probability that component k failed given system failure at time t is:

$$p_{\mathrm{K}|\mathrm{S}}(k|t, \Theta^\star) = \frac{f_k(t) \prod_{j \neq k} R_j(t)}{f_{\mathrm{S}}(t)} \tag{6}$$

Under the uniform masking model (Definition 2.3) with fixed cardinality w, we derive the conditional probability of observing candidate set C given system failure at time t.

Proposition 3.1. *Under uniform masking with cardinality w, the conditional probability of observing candidate set* C *given system failure at time t is:*

$$p_{\mathrm{C|S,W}}(\mathrm{C}|t, w, \Theta^\star) = \frac{\sum_{j \in \mathrm{C}} f_j(t) \prod_{k \neq j} R_k(t)}{\binom{m-1}{w-1} f_{\mathrm{S}}(t)} \tag{7}$$

Proof. By the law of total probability and independence of masking from system lifetime (Assumption 2.9):

$$\begin{aligned} p_{\mathrm{C|S}}(\mathrm{C}|t, w) &= \sum_{k=1}^{m} p_{\mathrm{C|K,S}}(\mathrm{C}|k, t, w) \cdot p_{\mathrm{K|S}}(k|t) \\ &= \sum_{k=1}^{m} p_{\mathrm{C|K}}(\mathrm{C}|k, w) \cdot p_{\mathrm{K|S}}(k|t) \end{aligned} \tag{8}$$

Under uniform masking (Definition 2.3):

$$p_{\mathrm{C|K}}(\mathrm{C}|k, w) = \begin{cases} \frac{1}{\binom{m-1}{w-1}} & \text{if } k \in \mathrm{C} \\ 0 & \text{if } k \notin \mathrm{C} \end{cases} \tag{9}$$

Therefore:

$$\begin{aligned} p_{\mathrm{C|S}}(\mathrm{C}|t, w) &= \sum_{k \in \mathrm{C}} \frac{1}{\binom{m-1}{w-1}} \cdot \frac{f_k(t) \prod_{j \neq k} R_j(t)}{f_{\mathrm{S}}(t)} \\ &= \frac{1}{\binom{m-1}{w-1}} \cdot \frac{\sum_{k \in \mathrm{C}} f_k(t) \prod_{j \neq k} R_j(t)}{f_{\mathrm{S}}(t)} \end{aligned} \tag{10}$$

□

This result shows that the candidate set probability is proportional to the sum of individual component failure probabilities over all components in the candidate set, normalized by the binomial coefficient representing the number of candidate sets containing each component.

The joint density of system failure time and candidate set is:

$$f_{\mathrm{C,S|W}}(\mathrm{C}, t|w, \Theta^\star) = \frac{1}{\binom{m-1}{w-1}} \sum_{j \in \mathrm{C}} f_j(t) \prod_{k \neq j} R_k(t) \tag{11}$$

The likelihood function for sample $\mathbf{M}_n$ is:

$$\mathcal{L}(\Theta|\mathbf{M}_n) = \prod_{i=1}^{n} f_{\mathrm{C,S|W}}(\mathrm{C}_i, \mathrm{t}_i|w, \Theta) \tag{12}$$

3.2 Fisher Information Matrix

The Fisher information matrix quantifies the expected information about $\Theta^\star$ contained in a single observation. Under regularity conditions, the (i, j)-th element is:

$$[\mathcal{I}(\Theta^\star|w)]_{ij} = -\mathrm{E}\{\frac{\partial^2}{\partial\theta_i\partial\theta_j}\ln f_{\mathrm{C,S|W}}(\mathrm{C},\mathrm{S}|w,\Theta)\}[\Theta^\star] \tag{13}$$

For a sample of n observations, the information is additive:

$$\mathcal{I}_n(\Theta^\star|w) = n\cdot\mathcal{I}(\Theta^\star|w) \tag{14}$$

4 Exponentially Distributed Component Lifetimes

4.1 Exponential Parametric Functions

Suppose component j has an exponentially distributed lifetime with failure rate $\lambda_j^\star$, denoted $\mathrm{T}_j \sim \mathrm{EXP}(\lambda_j^\star)$. The component has:

$$R_j(t|\lambda_j^\star) = \exp(-\lambda_j^\star t) \tag{15}$$

$$f_j(t|\lambda_j^\star) = \lambda_j^\star\exp(-\lambda_j^\star t) \tag{16}$$

$$h_j(t|\lambda_j^\star) = \lambda_j^\star \tag{17}$$

where $t > 0$ and $\lambda_j^\star > 0$.

Theorem 4.1. *A series system with exponentially distributed component lifetimes is exponentially distributed with failure rate* $\sum_{j=1}^m \lambda_j^\star$.

Proof. By equation (2),

$$R_\mathrm{S}(t|\boldsymbol{\lambda}^\star) = \prod_{j=1}^m \exp(-\lambda_j^\star t) = \exp\left(-\left[\sum_{j=1}^m \lambda_j^\star\right]t\right) \tag{18}$$

which is the reliability function of an exponential distribution with rate $\sum_{j=1}^m \lambda_j^\star$. $\square$

The system PDF is:

$$f_{\mathrm{S}}(t|\boldsymbol{\lambda}^\star) = \left(\sum_{j=1}^{m} \lambda_j^\star\right) \exp\left(-\left[\sum_{j=1}^{m} \lambda_j^\star\right] t\right) \tag{19}$$

Proposition 4.2. *For exponential series systems, the failed component* K *and system lifetime* S *are independent. The marginal distribution of* K *is:*

$$p_{\mathrm{K}}(k|\boldsymbol{\lambda}^\star) = \frac{\lambda_k^\star}{\sum_{j=1}^{m} \lambda_j^\star} \tag{20}$$

Proof. Let $\Lambda = \sum_{j=1}^{m} \lambda_j^\star$. We compute:

$$\begin{aligned} \mathrm{P}\{\mathrm{K} = k, \mathrm{S} \leq t\} &= \int_0^t \lambda_k^\star \exp(-\Lambda s)\, ds = \frac{\lambda_k^\star}{\Lambda}\left(1 - e^{-\Lambda t}\right) \\ &= \mathrm{P}\{\mathrm{K} = k\} \cdot \mathrm{P}\{\mathrm{S} \leq t\} \end{aligned} \tag{21}$$

since $\mathrm{P}\{\mathrm{K} = k\} = \lambda_k^\star/\Lambda$ and $\mathrm{P}\{\mathrm{S} \leq t\} = 1 - e^{-\Lambda t}$. □

Remark 4.3. Intuitively, the memoryless property of exponential distributions means that at any instant, each component has the same probability of being the next to fail, proportional to its rate λ_j. The *when* (system lifetime) and *which* (failed component) are determined by independent processes. This independence is specific to exponential distributions; for Weibull or other distributions with time-varying hazard rates, K and S are generally dependent.

The joint density is:

$$f_{\mathrm{K,S}}(k, t|\boldsymbol{\lambda}^\star) = \lambda_k^\star \exp\left(-\left[\sum_{j=1}^{m} \lambda_j^\star\right] t\right) \tag{22}$$

Under the uniform masking model (Definition 2.3), the joint density of candidate set and system failure time is:

$$f_{\mathrm{C,S|W}}(\mathrm{C}, t|w, \boldsymbol{\lambda}^\star) = \frac{1}{\binom{m-1}{w-1}} \left(\sum_{j \in \mathrm{C}} \lambda_j^\star\right) \exp\left(-\left[\sum_{j=1}^{m} \lambda_j^\star\right] t\right) \tag{23}$$

This follows directly from Proposition 3.1 by substituting the exponential PDFs and noting that for exponential distributions,

$$\prod_{k \neq j} R_k(t) = \exp\left(-\sum_{k \neq j} \lambda_k^\star t\right).$$

4.2 Maximum Likelihood Estimator

The likelihood function for sample $\mathbf{M}_n$ with candidate sets of cardinality w is:

$$\mathcal{L}(\boldsymbol{\lambda}|\mathbf{M}_n) \propto \exp\left(-\left[\sum_{j=1}^{m}\lambda_j\right]\left[\sum_{i=1}^{n}\mathrm{t}_i\right]\right)\prod_{i=1}^{n}\left(\sum_{j\in \mathrm{C}_i}\lambda_j\right) \tag{24}$$

The log-likelihood is:

$$\ell(\boldsymbol{\lambda}|\mathbf{M}_n) = \sum_{i=1}^{n}\ln\left(\sum_{j\in \mathrm{C}_i}\lambda_j\right) - \left[\sum_{j=1}^{m}\lambda_j\right]\left[\sum_{i=1}^{n}\mathrm{t}_i\right] \tag{25}$$

The score function has j-th component:

$$\frac{\partial \ell}{\partial \lambda_j} = \sum_{i=1}^{n}\frac{\mathbb{1}_{\mathrm{C}_i}(j)}{\sum_{k\in \mathrm{C}_i}\lambda_k} - \sum_{i=1}^{n}\mathrm{t}_i \tag{26}$$

where $\mathbb{1}_{\mathrm{C}}(j)$ is the indicator function equal to 1 if $j \in \mathrm{C}$ and 0 otherwise.

Remark 4.4. For general masking cardinality w, the score equations (26) do not admit closed-form solutions and the MLE must be computed numerically; Appendix A provides a Newton–Raphson algorithm. However, for $w = m-1$ the score equations linearize and a closed-form solution exists (Theorem 5.1).

4.3 Sufficient Statistics

Theorem 4.5. *For masked system failure times from exponentially distributed series systems, the statistics*

$$\bar{t} = \frac{1}{n}\sum_{i=1}^{n}\mathrm{t}_i \quad \textit{and} \quad \hat{\boldsymbol{\omega}} = (\hat{\boldsymbol{\omega}}_{\mathrm{C}})_{\mathrm{C}} \tag{27}$$

where $\hat{\boldsymbol{\omega}}_{\mathrm{C}}$ denotes the count of observations with candidate set C*, for each of the $\binom{m}{w}$ possible candidate sets, are jointly sufficient for $\boldsymbol{\lambda}^\star$.*

Proof. The likelihood can be factored as:

$$\mathcal{L}(\boldsymbol{\lambda}|\mathbf{M}_n) = \exp\left(-n\bar{t}\sum_{j=1}^{m}\lambda_j\right)\prod_{\mathrm{C}}\left(\sum_{j\in \mathrm{C}}\lambda_j\right)^{\hat{\boldsymbol{\omega}}_{\mathrm{C}}} \tag{28}$$

which depends on the data only through $\bar{t}$ and $\hat{\boldsymbol{\omega}}$. By the factorization theorem, these are sufficient statistics. □

The sufficiency result shows that all information about $\boldsymbol{\lambda}^\star$ in the sample is captured by: (1) the average system lifetime, and (2) the frequencies of each candidate set. We do not establish minimality: the likelihood is not a regular exponential family in $\boldsymbol{\lambda}$ because the natural parameters involve $\ln(\sum_{j\in \mathrm{C}} \lambda_j)$, so a separate argument would be needed to show that no further reduction is possible.

4.4 Fisher Information Matrix

Proposition 4.6. *The (j,k)-th element of the Fisher information matrix for exponential series systems is:*

$$[\mathcal{I}(\boldsymbol{\lambda}^\star|w)]_{jk} = \frac{\sum_{\mathrm{C}} \left(\sum_{p\in \mathrm{C}} \lambda_p^\star\right)^{-1} \mathbb{1}_{\mathrm{C}\times\mathrm{C}}(j,k)}{\binom{m-1}{w-1} \sum_{p=1}^{m} \lambda_p^\star} \tag{29}$$

where the sum is over all candidate sets C *of cardinality w and $\mathbb{1}_{\mathrm{C}\times\mathrm{C}}(j,k) = \mathbb{1}_{\mathrm{C}}(j)\mathbb{1}_{\mathrm{C}}(k)$.*

This formula reveals that information about the pair (λ_j, λ_k) accrues only from candidate sets containing both components j and k (the indicator $\mathbb{1}_{\mathrm{C}\times\mathrm{C}}(j,k)$ ensures this). Each such candidate set contributes inversely to its total failure rate, so components with higher failure rates contribute less information per observation. The denominator scales with the total failure rate and the number of possible candidate sets.

Proof. From equation (23), the log-density is:

$$\ln f_{\mathrm{C,S|W}}(\mathrm{C}, t|w, \boldsymbol{\lambda}) = \ln\left(\sum_{p\in\mathrm{C}} \lambda_p\right) - \left(\sum_{q=1}^{m} \lambda_q\right) t + \text{const} \tag{30}$$

Taking the first partial derivative with respect to λ_j:

$$\frac{\partial}{\partial \lambda_j} \ln f_{\mathrm{C,S|W}} = \frac{\mathbb{1}_{\mathrm{C}}(j)}{\sum_{p\in\mathrm{C}} \lambda_p} - t \tag{31}$$

where $\mathbb{1}_{\mathrm{C}}(j) = 1$ if $j \in \mathrm{C}$ and 0 otherwise.

Taking the second partial derivative with respect to λ_k:

$$\begin{aligned}
\frac{\partial^2}{\partial\lambda_j\partial\lambda_k} \ln f_{\mathrm{C,S|W}} &= \frac{\partial}{\partial\lambda_k}\left[\frac{\mathbb{1}_{\mathrm{C}}(j)}{\sum_{p\in\mathrm{C}}\lambda_p}\right] \\
&= -\frac{\mathbb{1}_{\mathrm{C}}(j)\cdot\mathbb{1}_{\mathrm{C}}(k)}{(\sum_{p\in\mathrm{C}}\lambda_p)^2} \\
&= -\frac{\mathbb{1}_{\mathrm{C}\times\mathrm{C}}(j,k)}{(\sum_{p\in\mathrm{C}}\lambda_p)^2}
\end{aligned} \tag{32}$$

By definition of Fisher information:

$$\begin{aligned}
[\mathcal{I}(\boldsymbol{\lambda}^\star|w)]_{jk} &= -\mathrm{E}\{\frac{\partial^2}{\partial\lambda_j\partial\lambda_k}\ln f_{\mathrm{C,S|W}}\}[\boldsymbol{\lambda}^\star] \\
&= \sum_{\mathrm{C}:|\mathrm{C}|=w}\int_0^\infty \frac{\mathbb{1}_{\mathrm{C}\times\mathrm{C}}(j,k)}{(\sum_{p\in\mathrm{C}}\lambda_p^\star)^2}\cdot f_{\mathrm{C,S|W}}(\mathrm{C},t|w,\boldsymbol{\lambda}^\star)\,dt
\end{aligned} \tag{33}$$

Substituting the joint density from equation (23):

$$\begin{aligned}
[\mathcal{I}(\boldsymbol{\lambda}^\star|w)]_{jk} = \sum_{\mathrm{C}:|\mathrm{C}|=w} &\frac{\mathbb{1}_{\mathrm{C}\times\mathrm{C}}(j,k)}{(\sum_{p\in\mathrm{C}}\lambda_p^\star)^2}\cdot\frac{1}{\binom{m-1}{w-1}} \\
&\times\int_0^\infty\left(\sum_{p\in\mathrm{C}}\lambda_p^\star\right)\exp\left(-\left[\sum_{q=1}^m\lambda_q^\star\right]t\right)dt
\end{aligned} \tag{34}$$

Evaluating the integral using $\int_0^\infty e^{-at}dt = 1/a$ for $a > 0$:

$$\int_0^\infty\left(\sum_{p\in\mathrm{C}}\lambda_p^\star\right)\exp\left(-\left[\sum_{q=1}^m\lambda_q^\star\right]t\right)dt = \frac{\sum_{p\in\mathrm{C}}\lambda_p^\star}{\sum_{q=1}^m\lambda_q^\star} \tag{35}$$

Therefore:

$$\begin{aligned}
[\mathcal{I}(\boldsymbol{\lambda}^\star|w)]_{jk} &= \sum_{\mathrm{C}:|\mathrm{C}|=w}\frac{\mathbb{1}_{\mathrm{C}\times\mathrm{C}}(j,k)}{(\sum_{p\in\mathrm{C}}\lambda_p^\star)^2}\cdot\frac{1}{\binom{m-1}{w-1}}\cdot\frac{\sum_{p\in\mathrm{C}}\lambda_p^\star}{\sum_{q=1}^m\lambda_q^\star} \\
&= \frac{1}{\binom{m-1}{w-1}\sum_{q=1}^m\lambda_q^\star}\sum_{\mathrm{C}:|\mathrm{C}|=w}\frac{\mathbb{1}_{\mathrm{C}\times\mathrm{C}}(j,k)}{\sum_{p\in\mathrm{C}}\lambda_p^\star}
\end{aligned} \tag{36}$$

This completes the derivation of equation (29). □

Proposition 4.7 (Information Recovery under Deterministic Masking)**.** *For an m-component exponential series system with $1 \le w \le m-1$, suppose the masking mechanism is* separating-deterministic*: there exists an injection $\varphi\colon \{1,\ldots,m\} \to \{\mathrm{C} \subseteq \{1,\ldots,m\} : |\mathrm{C}| = w\}$ satisfying $k \in \varphi(k)$ and $\mathrm{P}\{\mathrm{C} = \varphi(k) \mid \mathrm{K} = k\} = 1$ for each k. Then the Fisher information matrix equals that of complete identification ($w = 1$):*

$$[\mathcal{I}(\boldsymbol{\lambda}^\star \mid \varphi)]_{jk} = \frac{\delta_{jk}}{\Lambda\,\lambda_j^\star} \tag{37}$$

where $\Lambda = \sum_{j=1}^m \lambda_j^\star$ and δ_{jk} is the Kronecker delta. In particular, for $w = m-1$ the candidate sets $\{1,\ldots,m\}\setminus\{j\}$ are in bijection with the m components, and any derangement σ (a permutation with no fixed point) defines such a mechanism via $\varphi(k) = \{1,\ldots,m\} \setminus \{\sigma(k)\}$. Derangements exist for all $m \ge 2$.

Proof. Since φ is injective, each candidate set in the image of φ is produced by exactly one component, so the map $\mathrm{C} \mapsto \varphi^{-1}(\mathrm{C})$ recovers the failed component with certainty. Marginalizing the joint density (22) over the failure cause:

$$f(\mathrm{t}, \mathrm{C} \mid \boldsymbol{\lambda}) = \sum_{k=1}^m \lambda_k^\star\, e^{-\Lambda \mathrm{t}}\, \mathrm{P}\{\mathrm{C} = \mathrm{C} \mid \mathrm{K} = k\} = \lambda_{\varphi^{-1}(\mathrm{C})}\, e^{-\Lambda \mathrm{t}},$$

where the second equality uses $\mathrm{P}\{\mathrm{C} = \mathrm{C} \mid \mathrm{K} = k\} = \mathbb{1}_{\{\varphi(k)\}}(\mathrm{C})$ and the injectivity of φ. The log-density is thus

$$\ln f(\mathrm{t}, \mathrm{C} \mid \boldsymbol{\lambda}) = \ln \lambda_{\varphi^{-1}(\mathrm{C})} - \Lambda\,\mathrm{t},$$

which has the same functional form as the unmasked log-density with identified cause $\mathrm{K} = \varphi^{-1}(\mathrm{C})$. The second partial derivative is $\partial^2/\partial\lambda_j\partial\lambda_k = -\delta_{j,\varphi^{-1}(\mathrm{C})}\,\delta_{jk}/\lambda_j^2$, and taking the expectation over (S, C) gives $[\mathcal{I}]_{jk} = \delta_{jk}\,\mathrm{P}\{\mathrm{K} = j\}/\lambda_j^2 = \delta_{jk}/(\Lambda\,\lambda_j)$, where the last step uses $\mathrm{P}\{\mathrm{K} = j\} = \lambda_j/\Lambda$ (Proposition 4.2).

For the existence claim: when $w = m-1$, the candidate sets $c_j = \{1,\ldots,m\} \setminus \{j\}$ satisfy $k \in c_j$ if and only if $j \ne k$, so the map $\varphi(k) = c_{\sigma(k)}$ satisfies $k \in \varphi(k)$ whenever $\sigma(k) \ne k$. The number of derangements of $\{1,\ldots,m\}$ is $D_m = m! \sum_{i=0}^m (-1)^i/i! \ge 1$ for $m \ge 2$. □

Remark 4.8. Proposition 4.7 shows that the Fisher information loss from masking at cardinality w is not inherent to the cardinality, but depends on the masking mechanism. A well-designed diagnostic that always maps each failure cause to a unique candidate set incurs *no information loss whatsoever*, even at $w = m - 1$. The uniform masking model studied in this paper lies at the opposite extreme: it provides no preferential information about component identity, and the resulting FIM (Proposition 4.6) is strictly smaller than the complete-identification FIM. Note that the separating-deterministic mechanism violates the symmetry condition C2 (see Remark 2.8), since $\mathrm{P}\{\mathrm{C} = \varphi(k) \mid \mathrm{K} = k\} = 1$ but $\mathrm{P}\{\mathrm{C} = \varphi(k) \mid \mathrm{K} = k'\} = 0$ for $k' \neq k$ with $k' \in \varphi(k)$.

Remark 4.9 (Combining Estimates from Variable Masking Cardinality). In practice, diagnostic quality may vary across observations, resulting in data with different masking cardinalities. Suppose the sample is partitioned into G groups, where group g has n_g observations with masking cardinality w_g. Each group yields an MLE $\hat{\boldsymbol{\lambda}}^{(g)}$ with asymptotic covariance $n_g^{-1}\mathcal{I}^{-1}(\boldsymbol{\lambda}^\star|w_g)$.

The optimal combined estimator under independence is the inverse-variance weighted average:

$$\hat{\boldsymbol{\lambda}}_{\text{combined}} = \left(\sum_{g=1}^{G} n_g \mathcal{I}(\boldsymbol{\lambda}^\star|w_g)\right)^{-1} \times \left(\sum_{g=1}^{G} n_g \mathcal{I}(\boldsymbol{\lambda}^\star|w_g)\hat{\boldsymbol{\lambda}}^{(g)}\right) \tag{38}$$

with asymptotic covariance

$$\operatorname{Cov}(\hat{\boldsymbol{\lambda}}_{\text{combined}}) = \left(\sum_{g=1}^{G} n_g \mathcal{I}(\boldsymbol{\lambda}^\star|w_g)\right)^{-1} \tag{39}$$

This combined estimator achieves minimum variance among all linear unbiased combinations. In practice, the true parameter $\boldsymbol{\lambda}^\star$ in the information matrices is replaced by the combined estimate $\hat{\boldsymbol{\lambda}}_{\text{combined}}$, yielding a feasible estimator.

4.5 Asymptotic Sampling Distribution

The following regularity conditions ensure consistency and asymptotic normality of the MLE:

(R1) The true parameter $\boldsymbol{\lambda}^\star$ lies in the interior of the parameter space $\Theta = (0, \infty)^m$

(R2) The Fisher information matrix $\mathcal{I}(\boldsymbol{\lambda}^\star|w)$ is positive definite

(R3) The observations $(\mathrm{t}_i, \mathrm{C}_i)_{i=1}^n$ are independent and identically distributed

Condition (R1) excludes boundary cases where components have zero failure rate. Condition (R2) is satisfied when $1 \leq w < m$: from Proposition 4.6, the FIM decomposes as

$$\mathcal{I}(\boldsymbol{\lambda}^\star|w) = \frac{1}{\binom{m-1}{w-1}\Lambda} \sum_{\mathrm{C}:|\mathrm{C}|=w} \frac{\boldsymbol{e}_\mathrm{C}\boldsymbol{e}_\mathrm{C}^\top}{\sum_{p\in\mathrm{C}} \lambda_p^\star} \tag{40}$$

where $\boldsymbol{e}_\mathrm{C} \in \{0,1\}^m$ is the indicator vector of C and $\Lambda = \sum_j \lambda_j^\star$. Each summand is positive semidefinite, so for any $\boldsymbol{v} \neq \mathbf{0}$ it suffices to show $\boldsymbol{e}_\mathrm{C}^\top \boldsymbol{v} \neq 0$ for some C. Suppose $\boldsymbol{e}_\mathrm{C}^\top \boldsymbol{v} = 0$ for all $|\mathrm{C}| = w$. For $w = 1$, the candidate sets are singletons $\{j\}$, so $v_j = 0$ for each j and $\boldsymbol{v} = \mathbf{0}$, a contradiction. For $w \geq 2$, any two components $j \neq k$ appear together in some size-w subset; replacing one member by the other in such a set and subtracting gives $v_j = v_k$. Hence all entries of $\boldsymbol{v}$ are equal, and $wv_1 = 0$ forces $\boldsymbol{v} = \mathbf{0}$, a contradiction. At the boundary $w = m$, all observations share the single candidate set $\{1, \ldots, m\}$, the FIM has rank one, and only the total rate $\sum_j \lambda_j$ is identifiable. Condition (R3) holds by the experimental design.

Theorem 4.10. *Under conditions (R1)–(R3), as $n \to \infty$,*

$$\sqrt{n}(\hat{\boldsymbol{\lambda}}_n - \boldsymbol{\lambda}^\star) \xrightarrow{d} \mathrm{MVN}\left(\mathbf{0}, \mathcal{I}^{-1}(\boldsymbol{\lambda}^\star|w)\right) \tag{41}$$

This result follows from standard maximum likelihood theory [3, Theorem 2.5.1]. The asymptotic variance-covariance matrix is the inverse of the Fisher information matrix.

4.6 Confidence Intervals

An asymptotic $(1-\alpha) \times 100\%$ confidence interval for $\lambda_j^\star$ is:

$$\hat{\lambda}_j \pm z_{1-\alpha/2}\sqrt{\frac{1}{n}[\mathcal{I}^{-1}(\hat{\boldsymbol{\lambda}}_n|w)]_{jj}} \tag{42}$$

where $z_{1-\alpha/2}$ is the $(1-\alpha/2)$-quantile of the standard normal distribution.

5 Explicit MLE for $w = m - 1$

We derive the closed-form MLE for general m-component systems with masking cardinality $w = m - 1$, and then develop the three-component case ($m = 3$, $w = 2$) in detail as an illustration.

5.1 Closed-Form MLE

When each candidate set has cardinality $w = m - 1$, exactly one component is excluded from each candidate set. The score equations linearize, yielding a closed-form solution for arbitrary m.

Theorem 5.1. *For m-component systems with masking cardinality $w = m - 1$, define for each component j:*

- $A_j = \sum_{\mathrm{C} \ni j} \hat{\boldsymbol{\omega}}_{\mathrm{C}}$*: the sum of candidate set counts containing component j*
- $B_j = \hat{\boldsymbol{\omega}}_{\{1,\ldots,m\}\setminus\{j\}}$*: the count for the unique candidate set excluding component j*

The MLE has the closed-form solution:

$$\hat{\lambda}_j = \frac{A_j - (m-2)B_j}{n\bar{t}} \tag{43}$$

Proof. When $w = m-1$, there are exactly m candidate sets, each of size $m-1$, and each is the complement of a single component: $\mathrm{C}_{\setminus j} = \{1, \ldots, m\} \setminus \{j\}$ for $j = 1, \ldots, m$. Let $\sigma_{\setminus j} = \sum_{k \neq j} \lambda_k$ denote the sum of rates over $\mathrm{C}_{\setminus j}$.

From equation (26), the score equation for component j is:

$$\sum_{\substack{\mathrm{C}:|\mathrm{C}|=m-1 \\ j \in \mathrm{C}}} \frac{\hat{\boldsymbol{\omega}}_{\mathrm{C}}}{\sigma_{\mathrm{C}}} = n\bar{t} \tag{44}$$

where $\sigma_{\mathrm{C}} = \sum_{k \in \mathrm{C}} \lambda_k$. Component j appears in every candidate set except $\mathrm{C}_{\setminus j}$, so the sum ranges over $m-1$ terms.

Subtracting the score equation for component j' from that of component j: all candidate sets containing both j and j' cancel, leaving

$$\frac{\hat{\boldsymbol{\omega}}_{\mathrm{C}_{\setminus j'}}}{\sigma_{\setminus j'}} - \frac{\hat{\boldsymbol{\omega}}_{\mathrm{C}_{\setminus j}}}{\sigma_{\setminus j}} = 0 \quad \Rightarrow \quad \frac{\hat{\boldsymbol{\omega}}_{\mathrm{C}_{\setminus j'}}}{\sigma_{\setminus j'}} = \frac{\hat{\boldsymbol{\omega}}_{\mathrm{C}_{\setminus j}}}{\sigma_{\setminus j}} \tag{45}$$

for every pair j, j'. Thus there exists a constant γ such that $\sigma_{\setminus j} = \hat{\boldsymbol{\omega}}_{\mathrm{C}_{\setminus j}}/\gamma$ for all j. Substituting into any one score equation: each of the $m-1$ terms equals γ, so $(m-1)\gamma = n\bar{t}$, giving $\gamma = n\bar{t}/(m-1)$ and

$$\sigma_{\setminus j} = \frac{(m-1)\hat{\boldsymbol{\omega}}_{\mathrm{C}_{\setminus j}}}{n\bar{t}} \quad \text{for } j = 1, \ldots, m. \tag{46}$$

To recover the individual λ_j, note that $\sigma_{\setminus j} = \Lambda - \lambda_j$ where $\Lambda = \sum_{k=1}^{m} \lambda_k$. Summing equation (46) over j:

$$\sum_{j=1}^{m} \sigma_{\setminus j} = (m-1)\Lambda = \frac{(m-1)}{n\bar{t}} \sum_{j=1}^{m} \hat{\boldsymbol{\omega}}_{\mathrm{C}_{\setminus j}} = \frac{(m-1)n}{n\bar{t}} \tag{47}$$

since $\sum_j \hat{\boldsymbol{\omega}}_{\mathrm{C}_{\setminus j}} = n$. Therefore $\Lambda = 1/\bar{t}$. Then:

$$\hat{\lambda}_j = \Lambda - \sigma_{\setminus j} = \frac{1}{\bar{t}} - \frac{(m-1)\hat{\boldsymbol{\omega}}_{\mathrm{C}_{\setminus j}}}{n\bar{t}} = \frac{n-(m-1)B_j}{n\bar{t}} \tag{48}$$

where $B_j = \hat{\boldsymbol{\omega}}_{\mathrm{C}_{\setminus j}}$. Since $n = A_j + B_j$ (every observation either contains j or excludes it), we have $n-(m-1)B_j = A_j + B_j - (m-1)B_j = A_j - (m-2)B_j$, yielding equation (43). □

Remark 5.2. The closed-form solution exists because pairwise subtraction of score equations linearizes the system: all $(m-1)$-wise sums $\sigma_{\setminus j}$ are proportional to their candidate set counts. For $w < m-1$, each candidate set sum involves fewer than $m-1$ parameters, pairwise subtraction no longer isolates a single ratio, and the score equations do not admit this linearization. Closed-form solutions generally do not exist for $w < m-1$.

5.1.1 Illustration for Three Components

For $m = 3$ with $w = 2$, the candidate sets are $\{1,2\}$, $\{1,3\}$, and $\{2,3\}$, providing a concrete demonstration.

Corollary 5.3. *For three-component systems with $w = 2$, the MLE has the closed-form solution:*

$$\hat{\boldsymbol{\lambda}}_n = \frac{1}{n\bar{t}} \begin{pmatrix} \hat{\boldsymbol{\omega}}_{\{1,2\}} + \hat{\boldsymbol{\omega}}_{\{1,3\}} - \hat{\boldsymbol{\omega}}_{\{2,3\}} \\ \hat{\boldsymbol{\omega}}_{\{1,2\}} - \hat{\boldsymbol{\omega}}_{\{1,3\}} + \hat{\boldsymbol{\omega}}_{\{2,3\}} \\ -\hat{\boldsymbol{\omega}}_{\{1,2\}} + \hat{\boldsymbol{\omega}}_{\{1,3\}} + \hat{\boldsymbol{\omega}}_{\{2,3\}} \end{pmatrix}. \tag{49}$$

Proof. For $m = 3$: $A_1 = \hat{\boldsymbol{\omega}}_{\{1,2\}} + \hat{\boldsymbol{\omega}}_{\{1,3\}}$, $B_1 = \hat{\boldsymbol{\omega}}_{\{2,3\}}$, and $(m-2) = 1$. Applying Theorem 5.1: $\hat{\lambda}_1 = (A_1 - B_1)/(n\bar{t}) = (\hat{\boldsymbol{\omega}}_{\{1,2\}} + \hat{\boldsymbol{\omega}}_{\{1,3\}} - \hat{\boldsymbol{\omega}}_{\{2,3\}})/(n\bar{t})$. The formulas for $\hat{\lambda}_2$ and $\hat{\lambda}_3$ follow symmetrically. □

The log-likelihood for the three-component case is:

$$\begin{aligned} \ell(\boldsymbol{\lambda}|\bar{t}, \hat{\boldsymbol{\omega}}) = \hat{\boldsymbol{\omega}}_{\{1,2\}} \ln(\lambda_1 + \lambda_2) + \hat{\boldsymbol{\omega}}_{\{1,3\}} \ln(\lambda_1 + \lambda_3) \\ + \hat{\boldsymbol{\omega}}_{\{2,3\}} \ln(\lambda_2 + \lambda_3) - n\bar{t}(\lambda_1 + \lambda_2 + \lambda_3) \end{aligned} \tag{50}$$

Remark 5.4 (Boundary Estimates and the Triangle Inequality)*.* The closed-form MLE is the unconstrained solution to the score equations. From (49), $\hat{\lambda}_j > 0$ for all j if and only if the candidate set counts satisfy the *triangle inequality*:

$$\hat{\boldsymbol{\omega}}_{\{\mathrm{C}_{\setminus j}\}} < \sum_{j' \neq j} \hat{\boldsymbol{\omega}}_{\{\mathrm{C}_{\setminus j'}\}} \quad \text{for each } j = 1, 2, 3. \tag{51}$$

For example, $\hat{\lambda}_1 > 0$ requires $\hat{\boldsymbol{\omega}}_{\{2,3\}} < \hat{\boldsymbol{\omega}}_{\{1,2\}} + \hat{\boldsymbol{\omega}}_{\{1,3\}}$. This is the sample-level analogue of the separability condition [19]: if the candidate set counts satisfy the triangle inequality, the observed data separate all component pairs.

Under the uniform masking model, $\mathrm{P}\{\hat{\lambda}_j < 0\} \to 0$ as $n \to \infty$ when $\lambda_j^\star > 0$. However, for finite n the violation probability can be substantial, particularly when the failure rates are strongly asymmetric. Monte Carlo experiments (Section 5.3) show rejection rates of 3% at $n = 50$ for the symmetric configuration $\boldsymbol{\lambda}^\star = (3, 3, 3)^\top$ but 27% for the strongly asymmetric $\boldsymbol{\lambda}^\star = (1, 3, 5)^\top$. Negative estimates indicate:

1. Insufficient sample size for the degree of parameter asymmetry
2. True parameter near zero relative to other components
3. Potential model misspecification (uniform masking assumption violated)

When negative estimates occur, the constrained MLE (setting $\lambda_j = 0$ and re-solving for remaining components) also has closed form. For example, with $\lambda_1 = 0$:

$$\hat{\lambda}_2 = \frac{\hat{\boldsymbol{\omega}}_{\{1,2\}}}{(\hat{\boldsymbol{\omega}}_{\{1,2\}} + \hat{\boldsymbol{\omega}}_{\{1,3\}})\bar{t}}, \quad \hat{\lambda}_3 = \frac{\hat{\boldsymbol{\omega}}_{\{1,3\}}}{(\hat{\boldsymbol{\omega}}_{\{1,2\}} + \hat{\boldsymbol{\omega}}_{\{1,3\}})\bar{t}} \tag{52}$$

This reduces to the two-component MLE for components 2 and 3, treating $\lambda_1 = 0$ as fixed.

The Fisher information matrix is:

$$\mathcal{I}(\boldsymbol{\lambda}^\star|w{=}2) = \frac{1}{2(\lambda_1^\star + \lambda_2^\star + \lambda_3^\star)} \begin{bmatrix} \frac{1}{\lambda_1^\star+\lambda_2^\star} + \frac{1}{\lambda_1^\star+\lambda_3^\star} & \frac{1}{\lambda_1^\star+\lambda_2^\star} & \frac{1}{\lambda_1^\star+\lambda_3^\star} \\ \frac{1}{\lambda_1^\star+\lambda_2^\star} & \frac{1}{\lambda_1^\star+\lambda_2^\star} + \frac{1}{\lambda_2^\star+\lambda_3^\star} & \frac{1}{\lambda_2^\star+\lambda_3^\star} \\ \frac{1}{\lambda_1^\star+\lambda_3^\star} & \frac{1}{\lambda_2^\star+\lambda_3^\star} & \frac{1}{\lambda_1^\star+\lambda_3^\star} + \frac{1}{\lambda_2^\star+\lambda_3^\star} \end{bmatrix} \tag{53}$$

The inverse (asymptotic variance-covariance) is obtained by symbolic matrix inversion:

$$\mathcal{I}^{-1}(\boldsymbol{\lambda}^\star|w = 2) = (\lambda_1^\star + \lambda_2^\star + \lambda_3^\star) \begin{bmatrix} \lambda_1^\star + \lambda_2^\star + \lambda_3^\star & -\lambda_3^\star & -\lambda_2^\star \\ -\lambda_3^\star & \lambda_1^\star + \lambda_2^\star + \lambda_3^\star & -\lambda_1^\star \\ -\lambda_2^\star & -\lambda_1^\star & \lambda_1^\star + \lambda_2^\star + \lambda_3^\star \end{bmatrix} \tag{54}$$

This can be verified by direct multiplication: $\mathcal{I} \cdot \mathcal{I}^{-1} = \mathbf{I}_3$.

The asymptotic mean squared error (trace of variance-covariance) is:

$$\text{MSE}(\hat{\boldsymbol{\lambda}}_n) = \frac{3(\lambda_1^\star + \lambda_2^\star + \lambda_3^\star)^2}{n} \tag{55}$$

5.2 Candidate Sets of Size One

When $w = 1$, each observation identifies the exact failed component. This represents the no-masking case. The MLE is:

$$\hat{\boldsymbol{\lambda}}_n = \frac{1}{n\bar{t}} \begin{pmatrix} \hat{\boldsymbol{\omega}}_{\{1\}} \\ \hat{\boldsymbol{\omega}}_{\{2\}} \\ \hat{\boldsymbol{\omega}}_{\{3\}} \end{pmatrix} \tag{56}$$

The Fisher information matrix is diagonal:

$$\mathcal{I}(\boldsymbol{\lambda}^\star|w = 1) = \frac{1}{\lambda_1^\star + \lambda_2^\star + \lambda_3^\star} \text{diag}\left(\frac{1}{\lambda_1^\star}, \frac{1}{\lambda_2^\star}, \frac{1}{\lambda_3^\star}\right) \tag{57}$$

The asymptotic variance-covariance is:

$$\mathcal{I}^{-1}(\boldsymbol{\lambda}^\star|w=1) = (\lambda_1^\star + \lambda_2^\star + \lambda_3^\star)\text{diag}(\lambda_1^\star, \lambda_2^\star, \lambda_3^\star) \tag{58}$$

The MSE is:

$$\text{MSE}(\hat{\boldsymbol{\lambda}}_n|w=1) = \frac{(\lambda_1^\star + \lambda_2^\star + \lambda_3^\star)^2}{n} \tag{59}$$

which is exactly 1/3 the MSE when $w = 2$, reflecting the additional information from exact component identification.

Remark 5.5. The 1/3 MSE ratio holds regardless of the parameter values $\boldsymbol{\lambda}^\star$ because both MSE expressions factor as $(\sum_j \lambda_j^\star)^2/n$ times a constant that depends only on w, not on the individual $\lambda_j^\star$. This result is specific to the three-component system; for general m and w, the MSE ratio depends on both system dimension and masking cardinality.

5.3 Numerical Validation

We validate the asymptotic theory through Monte Carlo simulation studies.

5.3.1 Design

For each of three parameter configurations, we generate $R = 10^5$ independent samples of size n under the uniform masking model ($m = 3$, $w = 2$), compute the closed-form MLE via Corollary 5.3, and compare the empirical covariance of the MLEs with the theoretical asymptotic covariance $\frac{1}{n}\mathcal{I}^{-1}$.

The configurations span increasing parameter asymmetry:

- **Symmetric**: $\boldsymbol{\lambda}^\star = (3, 3, 3)^\top$ (equal failure rates)
- **Moderate**: $\boldsymbol{\lambda}^\star = (2, 3, 4)^\top$ (1:1.5:2 ratio)
- **Strong**: $\boldsymbol{\lambda}^\star = (1, 3, 5)^\top$ (1:3:5 ratio)

Sample sizes range from $n = 10$ to $n = 5000$.

Identifiability filtering. Samples that violate the triangle inequality (51)—i.e., where any MLE component is non-positive—are discarded before computing the empirical covariance. These samples do not satisfy the sample-level separability condition and are outside the regime where the asymptotic theory applies.

5.3.2 Results

Agreement is measured by the relative Frobenius error

$$\left\| \widehat{\mathrm{Cov}} - \tfrac{1}{n}\mathcal{I}^{-1} \right\|_F \Big/ \left\| \tfrac{1}{n}\mathcal{I}^{-1} \right\|_F ,$$

plotted in Figure 1.

For the symmetric and moderate configurations, the relative error tracks the $O(n^{-1/2})$ reference line closely and falls below 2% by $n = 200$. The strong asymmetry configuration $(1, 3, 5)$ converges more slowly—the error remains above 15% for $n \leq 200$, reflecting the fact that the rarest component ($\lambda_1 = 1$) is the actual failure cause in only $\lambda_1/\Lambda = 1/9$ of observations. For all configurations, the error drops below 1% by $n = 1000$.

Rejection rates. The fraction of non-identifiable samples decreases rapidly with n but depends strongly on the parameter configuration. At $n = 50$: the symmetric case rejects 3%, the moderate case rejects 9%, and the strong case rejects 27%. By $n = 200$, the symmetric and moderate cases show essentially zero rejections, while the strong case still rejects 7%. These rates provide practitioners with guidance on the minimum sample sizes needed for reliable closed-form inference: roughly $n \geq 50$ for balanced parameters, $n \geq 200$ for moderate imbalance, and $n \geq 500$ for strongly asymmetric configurations.

5.3.3 Implementation Note

Detailed simulation code, including data generation, MLE computation via Corollary 5.3, and metric calculation, is available in the companion repository at `https://github.com/queelius/expo-masked-fim`. The simulations were implemented in R using only base packages; the self-contained script `research/validate_asymptotic_covariance.R` reproduces all results in this section.

6 Conclusion

We have derived closed-form statistical inference for exponential series systems under uniform masking—a tractable specialization of the distribution-agnostic C1–C2–C3 framework [19]. The uniform masking assumption is

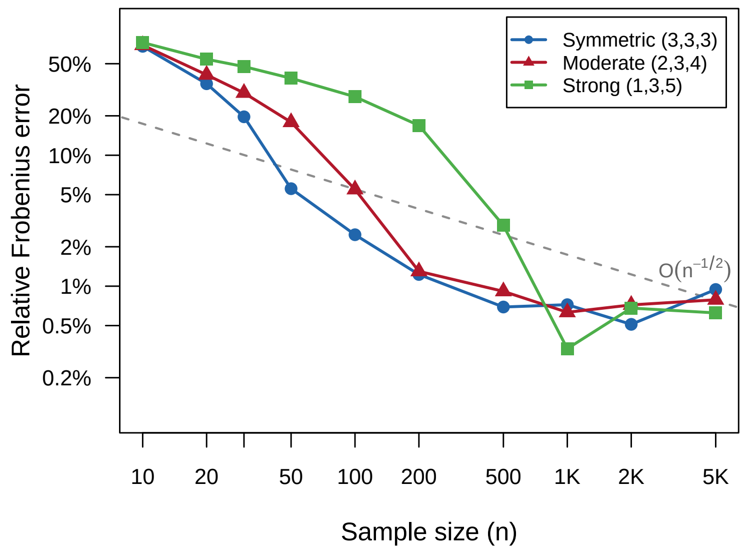

Figure 1: Relative Frobenius error between the empirical covariance of the MLE and the theoretical asymptotic covariance $\frac{1}{n}\mathcal{I}^{-1}(\boldsymbol{\lambda} \mid w{=}2)$, as a function of sample size n ($R = 10^5$ replications per point; non-identifiable samples filtered via the triangle inequality (51)). The symmetric and moderate configurations track the $O(n^{-1/2})$ reference line (dashed); the strongly asymmetric case converges more slowly due to the rarity of candidate sets involving the weakest component.

strong (Section 6.1), but it is precisely this strength that enables the complete analytical results obtained here.

The main insights are:

1. The Fisher information matrix has an explicit closed-form expression for any (m, w), quantifying how diagnostic precision (smaller w) translates into statistical efficiency

2. The mean system lifetime and candidate set frequency vector are sufficient statistics, reducing the data to $1 + \binom{m}{w}$ real numbers

3. For $w = m - 1$, the score equations linearize and a closed-form MLE exists for arbitrary m. Uniform masking at this cardinality minimizes the diagnostic mutual information I(K; C) among identifiable masking mechanisms (Proposition 2.6), providing a natural symmetry that en-

ables these analytical results. The closed-form FIM serves as a benchmark under maximal masking symmetry, enabling direct computation of asymptotic variances, confidence intervals, and sample size requirements

4. At $w = m$, the Fisher information matrix is singular, reflecting the non-identifiability of completely masked data established in the general theory [19]

5. Monte Carlo validation confirms that the asymptotic covariance matches empirical behavior for practical sample sizes (Figure 1)

6.1 Model Assumptions and Limitations

This work makes two key assumptions that limit its direct applicability but enable analytical tractability:

Uniform Masking Model: The assumption that non-failed components are equally likely to be included in the candidate set (Definition 2.3) is *not realistic in most practical scenarios*. Real diagnostic processes exhibit systematic biases:

- *Spatial clustering*: Components in the same physical location or subsystem are often grouped in candidate sets
- *Accessibility bias*: Easily inspected components may be overrepresented in candidate sets
- *Failure mode correlation*: Components with similar failure signatures are more likely to be co-nominated
- *Cost-based selection*: Expensive diagnostic tests may systematically exclude certain components

When the uniform masking assumption is violated, the likelihood in Section 3 is misspecified: the candidate-set term (7) is wrong. The resulting MLE converges to a *pseudo-true* parameter that solves the expected score under the incorrect masking model, which generally differs from the real parameter unless the non-uniform masking probabilities happen to be proportional across components. Intuitively, non-uniform masking up- or down-weights certain components in the log-likelihood, so the estimator absorbs masking bias instead of recovering the true failure rates. The Fisher information matrix derived here thus gives efficiency and variance only under uniform masking; with non-uniform masking, both bias and variance depend on the (unmodeled) masking mechanism.

Exponential Lifetimes: The constant-hazard assumption is appropriate for:

- Random external shocks (e.g., power surges, environmental failures)
- Systems without aging or wear-out (e.g., early-life failures)
- Components with memoryless failure processes

The exponential assumption fails when:

- Components exhibit wear-out (increasing hazard rate)
- Infant mortality dominates (decreasing hazard rate)
- Failure mechanisms are time-dependent

For non-exponential distributions (Weibull, gamma, lognormal), the likelihood equations generally do not admit closed-form solutions, and the Fisher information matrix must be computed numerically or via simulation; see [5, 14] for EM-based and Bayesian approaches that handle more general hazard structures.

Justification: Despite these limitations, we adopt these assumptions because they enable *complete analytical solutions*. The resulting formulas provide:

1. Baseline performance bounds for assessing more complex models
2. Insight into the fundamental information structure of masked data
3. Computational efficiency for quick parameter estimation
4. A reference model analogous to the role of the normal distribution in statistics

Practitioners should validate the uniform masking assumption by examining diagnostic process documentation or performing goodness-of-fit tests on candidate set patterns. When the assumption is questionable, simulation studies can assess robustness or alternative models (e.g., component-specific masking probabilities) should be considered.

6.2 Extensions

Several extensions would broaden the applicability of this framework:

1. **Non-uniform masking models**: Relax the uniform masking assumption to allow component-specific inclusion probabilities α_j or conditional dependence structures. For example:

   - Accessibility-weighted masking: $\mathrm{P}\{j \in \mathrm{C}|\mathrm{K} = k, j \neq k\} = \alpha_j$ depends on component accessibility
   - Cluster-based masking: Components within the same subsystem have correlated inclusion probabilities
   - Parametric masking models with estimable diagnostic parameters

   Such models sacrifice analytical tractability but more accurately reflect real diagnostic processes. The uniform masking results provide baseline comparisons and limiting cases.

2. **Variable masking cardinality**: Remark 4.9 provides the optimal combination formula when w varies across observations. Further extensions could model w as random, depending on diagnostic difficulty or component type

3. **Non-exponential lifetimes**: Extend to Weibull (aging/wear-out), gamma (multi-stage failures), or lognormal (time-dependent hazards) distributions. Numerical methods will be required, but asymptotic theory remains applicable.

4. **Covariate effects**: Incorporate covariate information (operating conditions, environmental factors) via proportional hazards models or accelerated failure time frameworks

5. **Bayesian approaches**: When prior information about component reliabilities is available, derive posterior distributions for $\boldsymbol{\lambda}$ and posterior predictive distributions for future failures

6. **Optimal diagnostic design**: Given costs of diagnostic tests and benefits of precise failure identification, design inspection strategies that optimize the information-cost tradeoff

The uniform masking, exponential lifetime framework provides a foundation for understanding masked system data. While restrictive, the analytical tractability enables clear insight into the information structure that extends conceptually to more complex settings. The closed-form results serve as benchmarks for evaluating numerical methods and assessing the efficiency loss from model misspecification.

A Numerical Solution for General w

Theorem 5.1 provides a closed-form MLE for $w = m - 1$, and the $w = 1$ case is trivial (exact component identification). For intermediate masking cardinalities $1 < w < m - 1$, the score equations (26) do not admit linearization and the MLE must be computed numerically. The Newton-Raphson algorithm is effective:

Algorithm 1: Newton-Raphson for Exponential MLE

Input: Initial guess $\boldsymbol{\lambda}^{(0)}$, tolerance ϵ
Output: MLE $\hat{\boldsymbol{\lambda}}_n$

$k \leftarrow 0$;
repeat
    Compute score $\boldsymbol{s}^{(k)} = \nabla \ell(\boldsymbol{\lambda}^{(k)} | \mathbf{M}_n)$;
    Compute Hessian $\mathbf{H}^{(k)} = \nabla^2 \ell(\boldsymbol{\lambda}^{(k)} | \mathbf{M}_n)$;
    Update $\boldsymbol{\lambda}^{(k+1)} = \boldsymbol{\lambda}^{(k)} - [\mathbf{H}^{(k)}]^{-1} \boldsymbol{s}^{(k)}$;
    $k \leftarrow k + 1$;
until $\|\boldsymbol{\lambda}^{(k+1)} - \boldsymbol{\lambda}^{(k)}\| < \epsilon$;
return $\boldsymbol{\lambda}^{(k)}$

The Hessian of the log-likelihood (note: negative semidefinite, so the observed information is $-\nabla^2 \ell$) is:

$$\left[\nabla^2 \ell(\boldsymbol{\lambda} | \mathbf{M}_n)\right]_{jk} = -\sum_{i=1}^{n} \frac{\mathbb{1}_{\mathrm{C}_i \times \mathrm{C}_i}(j, k)}{\left(\sum_{p \in \mathrm{C}_i} \lambda_p\right)^2} \tag{60}$$

Convergence is typically rapid when initialized at $\boldsymbol{\lambda}^{(0)} = \frac{1}{m\bar{t}}(1, \ldots, 1)$, which distributes the system rate $1/\bar{t}$ equally among components.

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